INCH-POUND

MIL-P-70995 (AR)
03 May 1991

MILITARY SPECIFICATION

PANEL, CONTROL, ARMAMENT SUBSYSTEM: M21

This specification is approved for use by the US Army Armament, Munitions and Chemical command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of control panel intended for use on the M21 Helicopter Armament Subsystem. The control panel consists of controls, switches, lamps, and electronic circuits necessary to synchronize gun position and gun firing with the sighting station.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, andhandbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Lead Alloy, Tin Lead Alloy and Tin Alloy; Flux cored Ribbon and Wire, and Solid Form

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1005

DISTRIBUTION STATEMENT A. Approved for public distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-W-13855	Weapons, Small Arms, General Specification for
MIL-W-16878	Wire, Electrical, insulated High Temperature
MIL-I-45607	inspection Equipment, Supply and Maintenance of
MIL-W-63150	Weapons and Support Materiel Standard, Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	- soldered Electrical and Electronic Assemblies
MIL-STD-45662	- Calibration Systems Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards,and handbooks are available from the Standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT, RESEARCH AND DEVELOPMENT AND ENGINEERING CENTER(ARDEC)

PRODUCT DRAWINGS

7792555	General Data - Electrical
7792127	Cover Assembly, Control Panel
7791891	Amplifier, Electronic Control - Control Panel
11699332	Panel, Control - Armament Subsystem
11699326	Panel, Edge Light - Control Panel

INSPECTION EQUIPMENT DraWINGS

IEL-1013172	Index of Inspection Equipment Lists
11016886	Setup and Operating Instructions for 11017474 Electrical Test Equipment
11017474	Test Equipment, Electrical
11018968	Gage, Digital Voltmeter-Radiometer

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

11018978	Test Equipment, Power Supply

PACKAGING DRAWINGS

P11699332	Packaging Data Sheet for Panel, Control Armament Subsystem

(Copies of drawings, publications and other Government documents required by manufactures in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and construction. control panels and parts shall conform to the materials and construction specified herein and on drawings referenced on Drawing 11699332 and shall be in accordance with the materials and construction provisions of MIL-W-13855.

3.2 Design. Control panels and parts shall conform to the design specified herein and on drawings referenced on Drawing 11699332 and shall be in accordance with the design provisions of MIL-W-13855.

3.2.1 Mechanical

3.2.1.1 Off-safe-armed switch The off-safe-armed switch shall be capable of being unlocked with a short pull-type motion without undue finger pressure and shall be retained in the off, safe, or armed position when released.

2.1.2 Gun selector switch. The gun selector switch shall be cable of being unlocked with a short pull type motion without undue finger pressure and shall be retained in the left, all or right posi- when released.

3.2.1.3 Amplifier access door. The amplifier access door shall be unlocked when the retaining stud is rotated counter-clockwise and shall be locked in the closed position when the stud is rotated clock wise. The unlocked door shall be capable of manual rotation to the full open (that which will permit amplifier installation and removal) and closed positions without binding or wobble.

3.2.1.4 Amplifiers. The amplifiers shall be securely positioned in the guide assembly and engaged in the connectors without relative movement and shall be removable under finger pressuur

3.2.1.5 Illuminating lamps The illuminating lamps shall be. securely retained in the plate assembly without relative movement and upon unscrewing shall be removable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

3.2.1.6 Press to test lamps. The press to test lamps shall move throughout the full range of travel when depressed and shall return to their initial position under spring action when released. The lamps shall be capable of assembling and disassembling without binding.

3.2.1.7 Edgelight panel and plate assembly. The edgelight panel and the plate assembly shall be securely retained to the forward panel so that there shall be no relative movement.

3.2.1.8 Cover assembly. The cover assembly shall be securely retained to the bracket assembly without relative movement and upon unlocking the retaining studs shall be removable.

3.2.1.9 Electrical connectors. The electrical connectors shall be securely retained in the bracket assembly so that there will be no relative movement.

3.2.1.10 Identification plate. The identification plate shall be securely attached to amplifier guide assembly so that there will be no relative movement.

3.2.2 Functioning. Control panels shall function properly when tested as specified in 4.5.4 utilizing a supply voltage of 24 to 30 volts direct current (v.d.c.).

3.2.3 Interchangeability. Unless otherwise specified on the drawings, all parts shall be interchangeable. (In normal assembly operations there shall be no objections interposed to preferential assembly of parts provided that all parts are dimensionally acceptable.) Control panels and repair parts shall be capable of meeting the interchangeability tests" specified in 4.5.5.

3.2.4 Reliability. The control panel shall have a reliable operating life of 60 test cycles without malfunction or unserviceable parts when tested as specified in 4.5.6.

3.2.5 Vibration. Control panels shall withstand vibration when tested as specified in 4.5.7.

3.3 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

3.4 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article as designated by the Contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Quantity
Control panel	Two

4.3.2 Inspections to be performed. (See Table I herein) As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70995 (AR)

PARAGRAPH	TITLE Control Panel	SHEET 1 OF 2	DRAWING NUMBER 11699332
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Materials and construction	COC	3.1	p4.5.2
	Mechanical	2 0-1	3.2.1	4.5.3
	Functioning	2 0-1	3.2.2	4.5.4
	Interchangeability	2 0-1	3.2.3	4.5.5
	Reliability	2 0-1	3.2.4	4.5.6
	Vibration	2 0-1	3.2.5	4.5.7
	Workmanship	2 0-1	3.3	Visual

NOTES:

Interpret Conformance Criteria column as follows:

COC indicates Certificate of Conformance

Number on left side of column indicates quantity to be tested

0-1 indicates pass on 0 defects, reject on 1 defect

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:17Z
Check the source to verify that this is the current version before use.

TABLE I First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Packaged Control Panel	2 OF 2	11699332
			NEXT HIGHER ASSEMBLY
			N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cleanliness	2 0-1	5.1	4.5.8.1
	Heat seal seam	2 0-1	5.1	4.5.8.2
	Vacuum retention	2 0-1	5.1	4.5.8.3
	Illegible or incorrect marking	2 0-1	5.1	Visual
	Improper location of marking	2 0-1	5.1	Visual
	Improper level of packaging and Packing	2 0-1	5.1	Visual
	Missing or inadequate quantity of Dessicant	2 0-1	5.1	Visual
	Humidity indicator missing	2 0-1	5.1	Visual
	Improper closure of interior packages	2 0-1	5.1	Visual
	Improper assembly of package: position of assemblies in supports and improper positioning of cushioning material	2 0-1	5.1	Visual
	Improper closure and strapping of shipping container	2 0-1	5.1	Visual

NOTES:
Interpret Conformance Criteria column as follows:
Number on left side of column indicates quantity to be tested
0-1 indicates pass on 0 defects, reject on 1 defect

Source: https://assist.dla.mil - Downloaded: 2026-02-20 20:04:17Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Unit of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, and same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor’s quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing, as a minimum, the conformance criteria specified. Where specified herein, attributes sampling inspection shall be conducted in accordance with Table II, using the inspection levels stated in the classification of Characteristics paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection

Lot Size	Level 1	Level 2	Level 3
2 to 8	*	5	3
9 to 15	13	5	3
16 to 25	13	5	3
26 to 50	13	5	3
51 to 90	13	12	4
91 to 150	13	12	5
151 to 280	30	14	6
281 to 500	30	17	7
501 to 1200	35	20	9
1201 to 3200	43	24	10
3201 to 10000	50	30	10
10001 to 35000	61	36	10
35001 to 150000	74	40	10
150001 to 500000	91	40	10
500001 and over	102	40	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection levels. Accept on zero and reject on one or more for all inspection levels.

4.4.2.2 Alternative quality conformance provisions. Alternative quality conformance-procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation. (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

PARAGRAPH 4.4.2.3	TITLE Control Panel	SHEET 1 OF 2	DRAWING NUMBER 11699332
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Materials and construction	COC	3.1	4.5.2
102	Mechanical	100% screening	3.2.1	4.5.3
103	Functioning	100% screening	3.2.2	4.5.4
104	Interchangeability	Level 3 0-1	3.2.3	4.5.5
105	Reliability	Level 3 0-1	3.2.4	4.5.6
106	Vibration	Level 3 0-1	3.2.5	4.5.7
Minor				
201	Workmanship	100% screening	3.3	Visual

NOTES:
Interpret conformance criteria column as follows:
COC indicates certificate of conformance
Level 3 is sample size as determined by Table II
0-1 indicates to accept lot on 0 defects, reject lot on 1 defect

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70995 (AR)

PARAGRAPH 4.4.2.3	TITLE Packaged Control Panel	SHEET 2 OF 2	DRAWING NUMBER 11699332
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Cleanliness	100% screening	5.1	4.5.8.1
102	Heat seal seam	Level 3 0-1	5.1	4.5.8.2
103	Vacuum retention	Level 3 0-1	5.1	4.5.8.3
Minor				
201	Illegible or incorrect marking	Level 1 0-1	5.1	Visual
202	Improper location of marking	Level 1 0-1	5.1	Visual
203	Improper level of packaging and Packing	Level 1 0-1	5.1	Visual
204	Missing or inadequate quantity of Dessicant	Level 2 0-1	5.1	Visual
205	Humidity indicator missing	Level 2 0-1	5.1	Visual
206	Improper closure of interior packages	Level 2 0-1	5.1	Visual
207	Improper assembly of package: position of assemblies in supports and improper positioning of cushioning material	Level 2 0-1	5.1	Visual
208	Improper closure and strapping of shipping container	Level 2 0-1	5.1	Visual

NOTES:
Interpret conformance criteria column as follows:
COC indicates certificate of conformance
Levels 1,2,and 3 are sample sizes as determined by Table II
0-1 indicates to accept lot on 0 defects, reject lot on 1 defect

Source: https://assist.dla.mil - Downloaded: 2026-05-22T04:17Z
Check the source to verify that this is the current version before use.

4.4.3 Inspection Equipment. Unless otherwise specified in procurement documents, responsibilities for acquisition, maintenance, and disposition of measuring and testing equipment prescribed on the Index of Inspection Equipment Lists, Drawing IEL-11013172, and for all other inspection equipment required to perform inspection prescribed herein and by other applicable specifications shall be in accordance with MIL-I-45607 and MIL-STD-45662 and MIL-W-63150. The contractor shall submit for approval Inspection equipment designs in accordance with the terms of the contract. (see 6.6)

4.5 Methods of inspection.

4.5.1 Test conditions.

4.5.1.1 Test equipment. Supply voltage, input signals and output loads of the helicopter armament subsystem are simulated by electrical test equipment in accordance with Drawing 11017474. The supply voltage for operation of the test equipment shall be 27 v.d.c.

4.5.1.2 Standard ambient conditions. Control panels shall be tested at a temperature of 60°F (15°C) to 90 F (32 C).

4.5.2 Materials and construction. Provide a Certificate of Conformance (COC) as defined in MIL-W-63150 for soldering (MIL-STD-2000), solder (QQ-S-571 and Drawing 11699332), and wire (MIL-W-16878 and Drawing 11699332).

4.5.3 Mechanical.

4.5.3.1 Off-safe-armed switch. Manually examine the off-safe-armed switch for compliance with 3.2.1.1.

4.5.3.2 Gun selector switch. Manually examine the gun selector switch for compliance with 3.2.1.2.

4.5.3.3 Amplifier access door. Manually examine the amplifier access door for compliance with 3.2.1.3.

4.5.3.4 Amplifiers. Visually and manually examine the amplifiers for compliance with 3.2.1.4.

4.5.3.5 Illuminating lamps. Manually examine the illuminating lamps for compliance with 3.2.1.6.

4.5.3.6 Press to test lamps. Manually examine the press to test lamps for compliance with 3.2.1.7.

4.5.3.7 Edgelight panel and plate assembly Visually and manually examine the edgelight panel for compliance with 3.2.1.8.

4.5.3.8 Cover assembly. Manually examine the cover assembly for compliance with 3.2.1.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

4.5.3.9 Electrical connectors. Manually examine the electrical connectors for compliance with 3.2.1.10.

4.5.3.10 Identification plate. Visually examine the identification plate for compliance with 3.2.1.11.

4.5.3.11 Wiring and sleeving installation. Visually examine the wiring and sleeving installation for compliance with drawing 7792555 and drawing 11699332.

4.5.4 Functioning. control panel shall be tested for functioning using set-up and operating instructions, gage, digital voltmeter-ratiometer, power supply, and electrical test equipment in accordance with drawings 11016886, 11018968, 11018978, and 11017474, respectively. Each control panel shall be tested twice for functioning, once utilizing 24 v.d.c. supply voltage and once utilizing 30 v.d.c supply voltage.

4.5.4.1 Lamps. The lamps shall be tested for proper functioning as follows:

a. Turn test equipment panel lamp switch on. To be acceptable, the panel lamps (DS 1003 and DS 1004) shall be illuminated.

b. Depress safe lamp (DS 1001) and armed lamp (DS 1002) of the control panel. To be acceptable, the lamps shall illuminate when depressed and extinguished when released.

4.5.4.2 Safe functioning. The control panel shall be tested for proper safe functioning as follows:

a. Move the off-safe-armed switch to the “safe” position. To be acceptable, the safe lamp on the control panel and the safe lamp on the test equipment shall illuminate. Rotate the safe lamp to each extreme position. To be acceptable, the safe lamp intensity shall change from dimmed at the full clockwise position and full brilliance at the full counter-clockwise position

b. Set amplifier selector of the test equipment at "left elevation". Read the voltage on the voltmeter. To be acceptable, the voltage shall be 21.6 v.d.c. ± 0.025 v.d.c. This shall be repeated for each of the four amplifier selections.

c. Set amplifier selector and follow-up variable resistor of the test equipment as indicated in Table III. Adjust the applicable stow variable resistor of the control panel to obtain a null on the nullmeter of the test equipment. To be acceptable, a null shall be obtained. This shall be performed for each of the four amplifier selecticons.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

Table III. Test equipment setting to test stow variable resistor.

Amplifier selector	Follow-up variable resistor
Left elevation	8500 divisions (div.)
Left deflection	8090 div.
Right elevation	1500 div.
Right deflection	8090 div.

d. Move the action switch to “on” and the amplifier selector to” "left elevation".Set the command variable resistor of the test equipment as specified in Table IV. Adjust the follow-up variable resistor of the test equipment to obtain a null on the nullmeter. To be acceptable, the reading on the follow-up variable resistor shall be in accordance with Table IV for each specified setting of the command variable resistor. This shall be performed for each of the four amplifier selections.

Table IV. Amplifier input and output signals.

Variable resistor of the test equipment	
Command	Follow-up
2000 div.	8000 div. ± 0050 div.
8000 div.	2000 div. ± 0050 div.
5000 div.	5000 div. ± 0050 div.

e. Move the action switch to “on”. Set the command variable resistor and the follow-up variable resistor as specified in Table V. read the current of servo signal A and servo signal C on the milliammeter of the test equipment. To be acceptable, the current of the servo signal A and servo signal C shall be in accordance with Table V. This shall be performed for each of the four amplifier selections.

Table V. Signal A and signal C current.

Variable resistor of the test equipment		Current (dc milliamperes)	
Follow-up	Command	Servo Signal A	Servo Signal C
5000 div.	5000 div.	4.5 to 6.0	4.5 to 6.0
5000 div.	0000 div.	1.5 maximum	7.5 minimum
5000 div.	9999 div.	7.5 minimum	1.5 maximum

4.5.4.3 Armed functioning. The control panel shall be tested for proper armed functioning as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

a. Move the off-safe-armed switch to the “armed” position. To be acceptable, the armed lamp on the control panel and the armed and safe lamps on the test equipment shall be illuminated; and the safe lamp On the control panel shall be extinguished. Rotate the armed lamp to each extreme position. To be acceptable, the armed lamp intensity shall change from dimmed at the full clockwise position to full brilliance at the full counter-clockwise position

b. Move the fire switch of the test equipment to “on”. Move the gun selector switch of the control panel to “all”. To be acceptable, the firing voltage lamps on the test equipment indicating firing voltage to all guns shall be illuminated.

c. Move the gun selector switch of the control panel to "right”. To be acceptable, the firing voltage lamps on the test equipment indicating firing voltage to the right gun shall be illuminated; and the firing voltage lamps on the test equipment indicating firing voltage to the left gun shall be extinguished.

d. Move the gun selector switch to the control panel to “left”. To be acceptable, the firing voltage lamps on the test equipment indicating firing voltage to the left gun shall be illuminated; and the firing voltage lamps on the test equipment indicating firing voltage to the right gun shall be extinguished.

4.5.5 Interchangeability. Control panels shall be tested for and shall comply with the requirements for functioning before and after this test method. Control panels shall be tested-for interchangeability by disassembling two control panels. This shall be accomplished by dividing the following listed parts into two different groups until each group contains a complete control panel. parts such as screws, nuts, washers, and pins shall be placed in the same group as their mating or associated part. Any screws, nuts, washers, and pins rendered unserviceable by disassembly shall be replaced without charge to the interchangeability test. The control panel shall be reassembled as specified in 4.5.5.2.

4.5.5.1 Disassembly.

4.5.5.1.1 Control panel 1. Disassemble control panel number 1 into the following groups:

Group 1A	Group 1B
Panel Edge Light - Control Panel-11699326; Cover Assembly, Control Panel - 7792127; Amplifier, Electronic Control - Control Panel - 7791891 (4 each)	All remaining parts.

4.5.5.1.2 Control panel 2. Disassemble control panel number 2 in the same manner as control panel 1, except number the groups 2A and 2B.

4.5.5.2 Assembly. Assemble control panels by combining groups in the following manner:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

Control Panel #1..... Group 1B and 2A
Control Panel #2..... Group 1A and 2B

4.5.6 Reliability. Control panels selected for the reliability test shall comply with the functioning requirement before and after the reliability test. The control panels shall be tested for reliability requirements under test conditions specified in 4.5.1.

4.5.6.1 Procedure.

a. Apply excitation supply voltages and loads
b. Depress safe and armed press to test lamps
c. Move the off-safe-armed switch to "safe" then to "armed".
d. Activate and deactivate the firing relay 24 times with the gun selector In “all”, 6 times in "right”, and 6 times with gun selector in “left”. Activation and deactivation duration shall be 10 seconds each.
e. Move the off-safe-armed switch to "off".
f. Repeat “a” through “e” 60 times.

4.5.6.2 Disposal. Reliabilty tested control panel shall be disposed of as specified in the contract (see 6.2).

4.5.7 Vibration. Control Panels shall be tested for vibration requirement in accordance with MIL-STD-810, Methods 514.4, procedure I, with test conditions In accordance with section I-3.4.3 and test set Up in accordance with section 11-1.2.1.4. Control panels shall be tested at levels defined by figure 514.4-9 and Table 514.4-IV of MIL-STD-810. The frequency of applied vibration shall be cycled logarithmically from 5 to 500 to 5 cycles per second for 5 cycles, each having a duration of 36 minutes, totaling 3 hrs. testing time for each axis. After vibration has been completed, the control panel shall be tested for and shall comply with the functioning requirement using the test method specified in 4.5.4. and shall be visually examined to determine that the requirements have been met. Vibration tested control panels shall be disposed of as specified in the contract.

4.5.8 Packaging tests.

4.5.8.1 Determination of cleanliness. The applicable surfaces of each sample unit shall be subjected to the determination of cleanliness test specified in MIL-P-116.

4.5.8.2 Heat seal seam. The same level A unit packages shall be subjected to the heat seal seam test specified in MIL-P--116.

4.5.8.3 Vacuum retention. A sufficient vacuum shall be drawn to cause the flexible barrier of unit packages to cling snugly to the enclosed item. Care shall be exercised to insure that an excessive amount of vacuum is not applied which might cause puncture or rupture of the barrier. Without releasing the vacuum, the final opening in the barrier shall be sealed. After remaining undisturbed at room tern perature for two hours, the barrier shall be. examined to determine

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

whether it is still taut and retracts against the item when drawn away and quickly released.

5. PACKAGING.

5.1 Preservation, packaging packing, and marking. Control panels shall be preserved, unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P11699332 for the level of protection specified (see 6.2). Each packaged control panel shall withstand the packaging tests of 4.5.8.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the assembling, testing, and packing for one type of control panel for use on the M21 Helicopter Armament Subsystem-

6.2 Acquisition Requirements. Acquisition documents should specify the following:

a. Title,number and date of this specification.
b. Lists of drawings and specifications pertinent to the control panel showing applicable revision dates.
c. inspection criteria for components (see 4.5).
d. Index of inspection equipment lists pertinent to the control panel, showing applicable revision dates.
e. That packages opened for examination shall be repackaged by the contractor at the contractor's expense.
f. Disposition of reliability and vibration tested control panels (see 4.5.7.2 and 4.5.8).
g. Number and list of parts required and shipping instructions for production samples.
h. Selection of applicable levels of preservation, packaging, and packing.
i. Packaging instructions for repair parts.
j. Place of final inspection and acceptance (see “Notes” in MIL-W-13855) .
k. List of acceptance inspection equipment to be furnished the contractor and responsibilities for other Government property to be furnished the contractor.
l. Quantity, shipping instructions, and test procedures for control panels parts when an interplant interchangeability test is reguired.
m. procedures and methods for demilitarizing and disposing of rejected material.
n. Disposition of Government furnished property.

6.3 Testing agency. When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

6.4 First Article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.4), and the number of items to be tested as specified in 4.3.1. The contracting officer should also include. specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Armament Reasearch and Development Command (ARRADCOM), or U.S. Army Armament Research and Development Drawings. Technical Data originally prepared by these activities are now under the cognizance of ARDEC.

6.6 Submission of inspection equipment designs for approval. Submit equipment designs as required to commander, ARDEC, ATTN: AMSMC-QAF-1, Picatinny Arsenal, NJ, 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.8 Submission of alternative inspection provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.7 Subject term (key words)listing

High Rate Gun
Rocket Launcher
Amplifier
Off-Safe-Armed Switch
Gun Selector Switch
Illuminating Lamps
Press to Test Lamps
Cover Assembly
Electrical Connectors
Wiring and Sleeving Installation

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1005-A751)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70995 (AR)	3 May 1991

3. DOCUMENT TITLE

PANEL, CONTROL, ARMAMENT SUBSYSTEM: M21

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: (201) 724-6662 (2) AUTOVON: 880-6662
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:17Z
Check the source to verify that this is the current version before use.